

Rupali Bhati

RESEARCH INTERESTS	Multi-Agent Reinforcement Learning, Cooperative AI, Reinforcement Learning from Human Feedback, Multi-Agent Systems, Game Theory
EDUCATION	Northeastern University (Supervisor: Christopher Amato) <i>Boston, U.S.A.</i> <i>Ph.D. in Computer Science</i> GPA: 3.93/4.0 <i>Sep 2023 - Present</i> Université Laval/ Mila (Supervisor: Audrey Durand) <i>Quebec, Canada</i> <i>M.Sc. in Computer Science (thesis)</i> GPA: 4.2/4.3 <i>Sep 2020 - Aug 2023</i> Delhi Technological University (Supervisor: Indu Sreedevi) <i>New Delhi, India</i> <i>B.Tech. in Electronics and Communication Engineering</i> Aggregate percentage: 72.29% (WES equivalent 3.55/4.0) <i>Aug 2012 - May 2016</i>
SELECTED PUBLICATIONS	- Decentralized Asymmetric DQN: Decentralization without Factorization in Multi-Agent Reinforcement Learning [link] Rupali Bhati*, Anurag Kadkol*, Andrea Baisero, Christopher Amato (*= equal contribution) <i>RLC 2026</i> - Fixing Incomplete Value Function Decomposition for Multi-Agent Reinforcement Learning [link] Andrea Baisero, Rupali Bhati, Shuo Liu, Aathira Sunil Pillai, Christopher Amato <i>RLC 2026</i> - The Influence of Scaffolds on Coordination Scaling Laws in LLM Agents [link] Rupali Bhati*, Mariana Meireles*, Niklas Lauffer[†], Cameron Allen[†] (*= equal contribution, [†] equal advising) <i>NeurIPS 2025 Scaling Environments for Agents Workshop</i> - On Stateful Value Factorization in Multi-Agent Reinforcement Learning [link] Enrico Marchesini, Andrea Baisero, Rupali Bhati, Christopher Amato <i>AAMAS 2025</i> - Curriculum Learning for Cooperation in Multi-Agent Reinforcement Learning [link] Rupali Bhati, SaiKrishna Gottipati, Clodéric Mars, Matthew E. Taylor <i>NeurIPS 2023 Agent Learning in Open-Endedness Workshop</i>
OTHER PUBLICATIONS	- Use of an Integrated Knowledge Translation Approach to Develop an Electronic Patient-Reported Outcome System for Cancer Rehabilitation: Tutorial [link] Christian Lopez, Sarah E Neil-Sztramko, Kristin L Campbell, David M Langelier, Tran Truong, Yuliya Gavrylyuk, Pia Nyakairu, Laura Parente, Audrey Durand, Jackie L Bender, Gillian Strudwick, Rupali Bhati, Jonathan Greenland, Tony Reiman, Jennifer M Jones <i>JMIR Cancer 2025</i> - Scalable Approaches for a Theory of Many Minds [link] Maximilian Puelma Touzel, Amin Memarian, Matthew Riemer, Andrei Mircea Romascanu, Andrew Williams, Elin Ahlstrand, Lucas Lehnert, Rupali Bhati, Guillaume Dumas, Irina Rish <i>ICML 2024 Agentic Markets Workshop</i> - Performative Prediction in Time Series: A Case Study [link] Rupali Bhati, Jennifer Jones, Kristin Campbell, David Langelier, Anthony Reiman, Jonathan Greenland, Audrey Durand <i>NeurIPS 2022 Workshop on Learning from Time Series for Health</i>

9. Summarizing Societies: Agent Abstraction in Multi-Agent Reinforcement Learning [\[link\]](#)
Amin Memarian, Maximilian Puelma Touzel, Matthew D Riemer, **Rupali Bhati**, Irina Rish
ICLR 2022 From Cells to Societies: Collective Learning across Scales Workshop
10. Interpret Your Care: Predicting the Evolution of Symptoms for Cancer Patients [\[link\]](#)
Rupali Bhati, Jennifer Jones, Audrey Durand
AAAI 2022 Trustworthy AI for Healthcare Workshop
11. CARL: Conditional-value-at-risk Adversarial Reinforcement Learning [\[link\]](#)
Mathieu Godbout, Maxime Heuillet, Sharath Chandra, **Rupali Bhati**, Audrey Durand
AAAI 2022 Safe AI Workshop
12. A Reinforcement Learning Approach to Jointly Adapt Vehicular Communications and Planning for Optimized Driving [\[link\]](#)
Mayank K. Pal, **Rupali Bhati**, Anil Sharma, Sanjit K. Kaul, Saket Anand, P.B.Sujit
IEEE ITSC 2018

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Intern, Center for Human Compatible AI (CHAI), UC Berkeley

Supervisor: Niklas Lauffer

May 2025 - Aug 2025

- Investigated how scaffolding affects cooperative scaling laws in multi-agent LLM systems.

Graduate Research Assistant, Northeastern University

Supervisor: Christopher Amato

Sep 2023 - Present

- Currently developing methods for reinforcement learning from human feedback (RLHF) in multi-agent systems, with a focus on learning cooperative behavior.
- Developed Dec-ADQN, a decentralized asymmetric Q-learning method that decouples decentralized execution from value factorization in cooperative multi-agent reinforcement learning.
- Investigated theoretical limitations of value decomposition methods and developed QFIX, an IGM-complete decomposition framework with a lightweight fixing layer, and DuelMIX, a stateful factorization method with distinct per-agent utility estimators.

Machine Learning Alignment & Theory (MATS) Scholar

Supervisor: Christian Schroeder de Witt

Jun 2024 - Aug 2024

- Explored the validity of the individual global max (IGM) principle during training in value decomposition methods for cooperative multi-agent reinforcement learning.

Research Intern

AI Redefined

Jan 2023 - Jul 2023

- Developed UED (unsupervised environment design)-inspired curriculum learning approaches to improve cooperation among reinforcement learning agents in the game of Overcooked.

Graduate Research Assistant, Université Laval & Mila

Supervisor: Audrey Durand

Sep 2020 - Aug 2023

- Addressed the problem of performative prediction in time-series data for predicting cancer-related fatigue and pain and successfully found stable points by applying repeated performative training.
- Formulated agent abstraction in the multi-agent setting and showed how it can help disentangle non-stationarity in the game of Diplomacy and achieve higher compression.

Reinforcement Learning Consultant

Self-Employed

Sep 2018 - Aug 2020

- At Bert Labs, applied RL to increase the energy efficiency of a HVAC system. For a leading global FMCG company's headquarters building, using DQN, increased efficiency of their Air-Handling Unit system by over 70% as compared to classical PID logic.
- Designed and delivered a week long workshop to teach fundamentals of RL to employees at Adventum. Consulted on application of RL to improve segmentation in medical images.
- Worked with CatapulZ to develop RL blue agents to Capture-The-Flag in cybersecurity applications.
- Worked at UpGrad as a Domain Expert to develop an end-to-end solution for a model inventory management problem to meet next-to-next day demand using DDQN.

Research Assistant, Indraprastha Institute of Information Technology - Delhi

Supervisor: Saket Anand

Sep 2017 - Aug 2018

- Trained an autonomous vehicle to smartly adapt communications and planning actions, while achieving large driving utilities using Q-learning.

Data Analyst

KPMG

Jun 2016 - Aug 2017

- In collaboration with Microsoft, developed a policy-iteration algorithm for dynamic ticket pricing to maximize revenue and reduce manual effort by 70-80%.
- Applied predictive and descriptive analytics to identify business insights and automation opportunities for government organizations.

SCHOLARSHIPS AND AWARDS

- 2023 Khoury Distinguished Fellowship.
- 2023 Awarded Sony Interactive Entertainment Scholarship to attend the Summer School on AI and Games. Awarded first place at Game AI Jam at the Summer School on AI and Games.
- 2022 **Google CSRMP**: Selected for Google Computer Science Research Mentorship Program with mentor Wenhao Yu.
- 2022 Second place at the Rendez-Vous IA Quebec.
- 2022 Institute of Intelligence and Data (IID) Laval Tuition Scholarship.
- 2022 Nominated for **Women in Artificial Intelligence Awards** North America.

TECHNICAL SKILLS

Python, JAX, JaxMARL, PyTorch, Reinforcement Learning, Multi-Agent Reinforcement Learning, LLMs, vLLM, Docker, Git

TEACHING EXPERIENCE

- **Head Teaching Assistant**, *CS 5180: Reinforcement Learning and Sequential Decision Making*, Northeastern University *Spring 2026*
- **Head Teaching Assistant**, *CS 5100: Foundations of AI*, Northeastern University *Fall 2025*
- **Teaching Assistant**, *GIF-7005: Introduction to ML*, Université Laval *Fall 2021*
- **Mentor**, Codementor [\[link\]](#) *Fall 2019 - Summer 2020*
- **Teaching Assistant**, Coding Blocks [\[link\]](#) *Summer 2018*
- **Teaching Assistant**, UpGrad *Fall 2018*

SERVICE

Organiser: Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI at ICLR 2025 [\[link\]](#)
Organiser: Coordination and Cooperation in Multi-Agent Reinforcement Learning (CoCoMARL) Workshop at The Reinforcement Learning Conference 2024 [\[link\]](#)
Organiser: Multi-Agent Learning Seminar [\[link\]](#)
Reviewer: EWRL 2026, Cooperative AI Summer School applications 2025, WiML Workshop at NeurIPS 2025, NeurIPS 2023, Montreal AI Symposium 2022, ITSC 2018
Facilitator: ICML WiML UnWorkshop: Machine Learning for Physical Sciences 2022
